

Practice Exam Three: Version One

Topics Covered:

- Properties of the Inductor and Capacitor
- RL/RC Circuits without Dependent Sources
- RL/RC Circuits with Dependent Sources

Practice Exam Tips:

- Treat this as timed practice, but do not rush. The goal is to identify gaps in understanding.
- After completing the exam, you can grade your work using the provided rubric in the solution. The practice solutions also include suggested discussion questions to try.
- If you make mistakes, revisit the relevant topic and rework the problem before reviewing the solution or attempting the second practice exam.
- Focus on method and clarity, not speed.

Exam Tips:

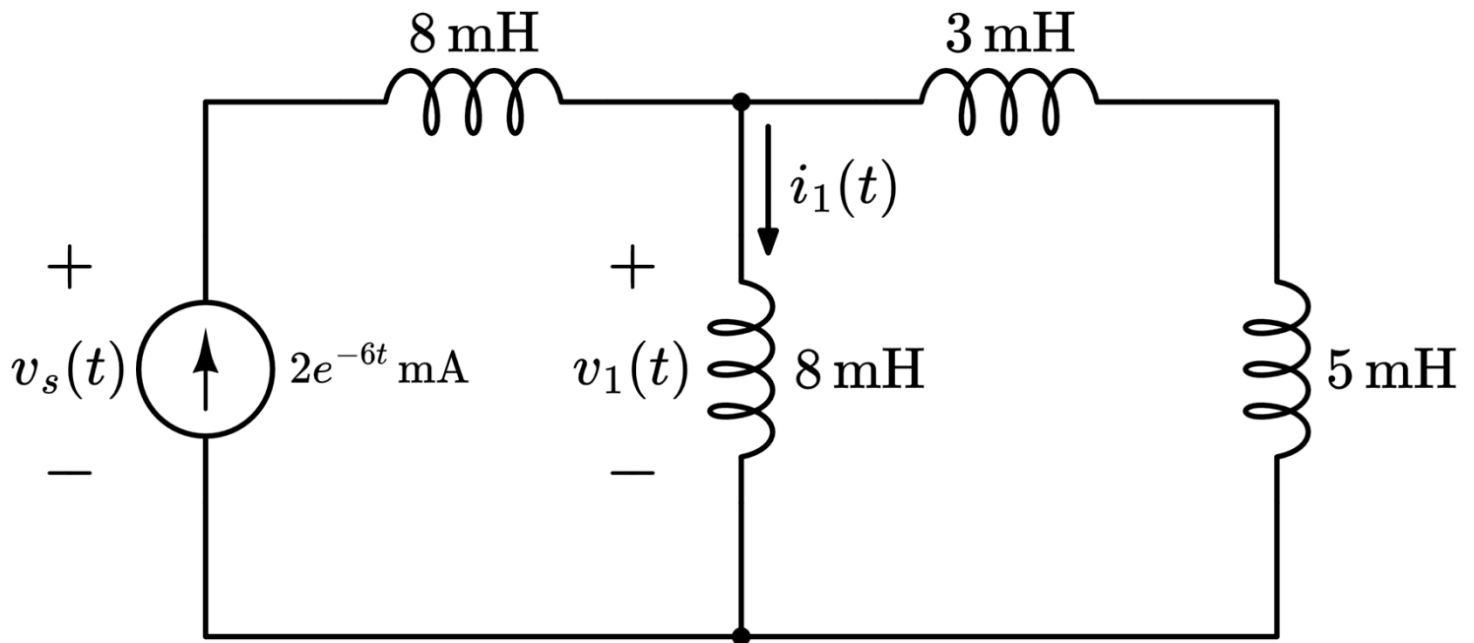
- Show all work clearly. Correct reasoning with minor algebra errors can still earn partial credit.
- Clearly label currents, voltages, and reference directions.
- Use consistent sign conventions throughout each problem.
- Include units in all final answers.
- The actual exam time limit is 75 minutes. If you follow the suggested pacing from this practice exam, you will finish in 60 minutes.

Question One: Properties of the Capacitor and Inductor

For the circuit shown below, find the following expressions for $t \geq 0$:

- a) $v_s(t)$
- b) $v_1(t)$
- c) $i_1(t)$

Suggested time limit: 20 minutes

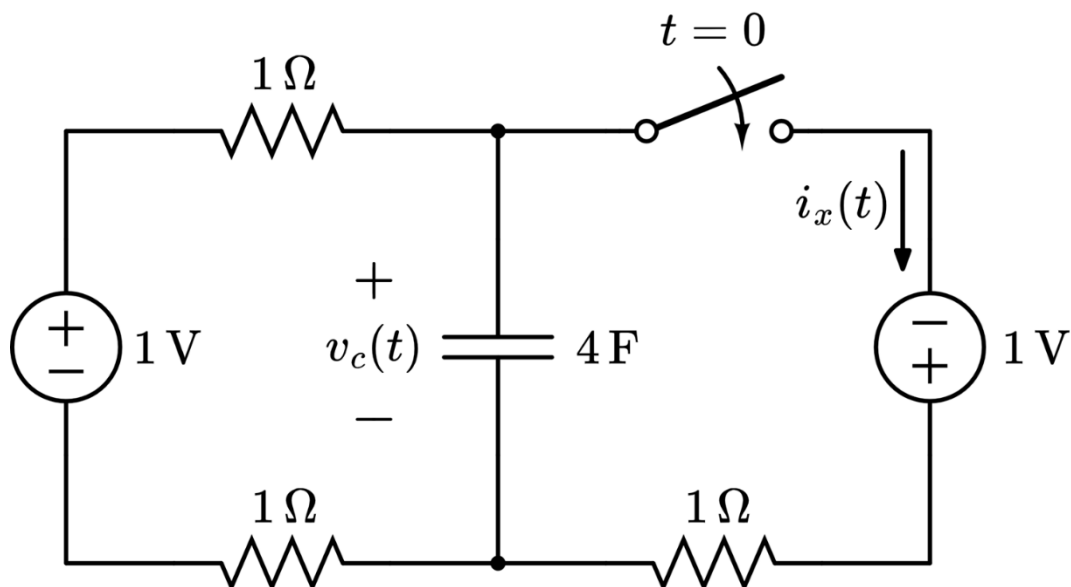


Question Two: RL/RC Circuits without Dependent Sources

For the circuit shown below, the switch has been open for a long time and closes at $t = 0$. Find the following expressions:

- a) $v_c(0)$
- b) $v_c(t)$
- c) $i_x(t)$

Suggested time limit: 20 minutes



Question Three: RL/RC Circuits with Dependent Sources

For the circuit shown below, the switch has been open for a long time and closes at $t = 0$. Find the following expressions:

- a) $i_L(0)$
- b) $i_L(t)$
- c) $i_x(t)$

Suggested time limit: 20 minutes

